

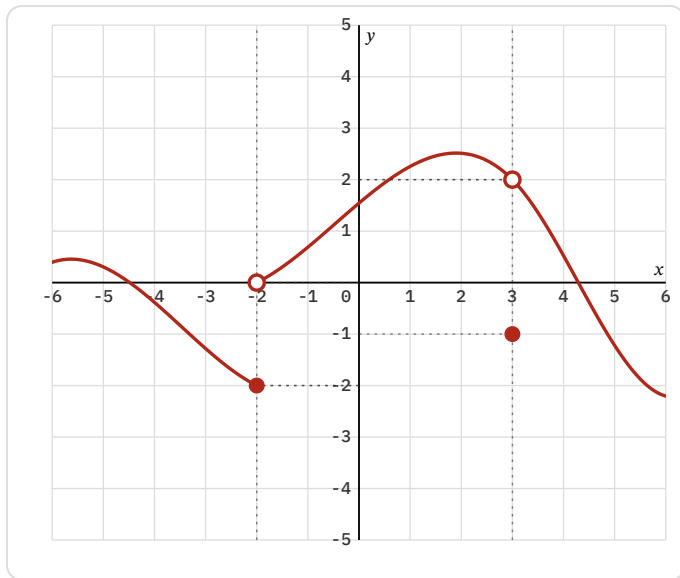
Mock Midterm B

Practice midterm (AI-Free) · Sections 1.1–1.4

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(-2)$ _____ (1)

b.) $\lim_{x \rightarrow -2^-} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow -2^+} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow -2} f(x)$ _____ (1)

e.) $f(3)$ _____ (1)

f.) $\lim_{x \rightarrow 3^-} f(x)$ _____ (1)

g.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (1)

h.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

8 pts

2. Let $f(x) = \frac{e^{-x}-1}{x}$. Use the table to see what the outputs do as x approaches 0.

7 pts

x	-0.5	-0.1	-0.01	0	0.01	0.1	0.5
$f(x)$	-1.297443	-1.051709	-1.005017	?	-0.995017	-0.951626	-0.786939

a.) Can you find $\lim_{x \rightarrow 0} f(x)$ just by plugging in $x = 0$? (1)

(A) Yes — $f(0)$ is defined, so substitute

(B) No — $x = 0$ makes a denominator 0, so $f(0)$ is undefined

b.) As $x \rightarrow 0^-$, $f(x)$ approaches... (2)

c.) As $x \rightarrow 0^+$, $f(x)$ approaches... (2)

d.) $\lim_{x \rightarrow 0} f(x) =$ (2)

3. Calculate each limit algebraically.

6 pts

a.) $\lim_{x \rightarrow 3} (-3x^2 + 4x + 5)$ (2)

b.) $\lim_{x \rightarrow 1} \frac{x^2-2x-1}{x-4}$ (2)

c.) $\lim_{x \rightarrow -4} \sqrt{x^2 + 33}$ (2)

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow -1} \frac{x+1}{x^2+2x+1}$ _____ (2)

b.) $\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h}$ _____ (2)

4 pts

5.

Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-4x^3+1}{4x^4-9x^2-9x-1}$ _____ (2)

b.) $\lim_{x \rightarrow -\infty} \frac{-2x^4+9x-5}{x^4+7x^3+7x-3}$ _____ (2)

c.) $\lim_{x \rightarrow \infty} \frac{7x^5-2x^4-5}{3x^3-4x^2-6x+5}$ _____ (2)

6.

Let

7 pts

$$G(x) = \begin{cases} x^2 - 3 & x < 0 \\ -2x & x > 0 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(0)$ _____ (1)

b.) $\lim_{x \rightarrow 0^-} G(x)$ _____ (1)

c.) $\lim_{x \rightarrow 0^+} G(x)$ _____ (1)

d.) $\lim_{x \rightarrow 0} G(x)$ _____ (1)

e.) $G(2)$ _____ (1)

f.) $\lim_{x \rightarrow -\infty} G(x)$ _____ (1)

g.) $\lim_{x \rightarrow \infty} G(x)$ _____ (1)

7.

Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 - 3x + 1}{2x - 1} & x < \frac{1}{2} \\ 2x - 2 & x > \frac{1}{2} \end{cases}$$

Find each limit or function value algebraically. Write **DNE** if it doesn't exist.

a.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -3} f(x)$ _____ (1)

c.) $f\left(\frac{1}{2}\right)$ _____ (1)

d.) $\lim_{x \rightarrow \frac{1}{2}^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow \frac{1}{2}^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow \frac{1}{2}} f(x)$ _____ (1)

8.

Find the value of k that makes f continuous at $x = 1$.

4 pts

$$f(x) = \begin{cases} x^2 & x < 1 \\ kx + 5 & x \geq 1 \end{cases}$$

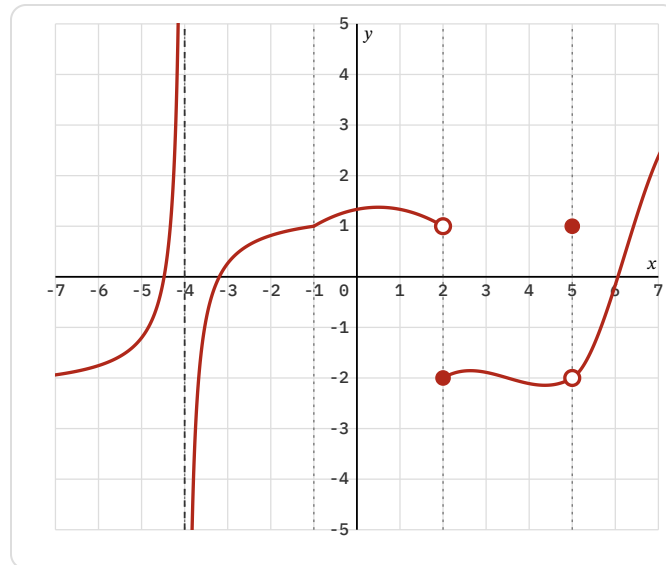
a.) $\lim_{x \rightarrow 1^-} f(x)$ _____ (2)

b.) $k =$ _____ (2)

9.

Classify the behavior of f at each labeled x -value.

4 pts

a.) At $x = -4$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

b.) At $x = -1$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

c.) At $x = 2$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

d.) At $x = 5$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

10.

Let $P(t) = 2t^2 + 4t + 56$, where $P(t)$ represents the population of a town, in thousands of people, t years after 2020. Find the average rate of change over $[1, 4]$.

a.) $P(1)$ _____ (2)

b.) $P(4)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

d.) Which sentence correctly states the result **with units**? (1)

(A) Between $t = 1$ and $t = 4$, the population changes at an average rate of **14** years per thousand people.

(B) Between $t = 1$ and $t = 4$ years, the population changes at an average rate of **14** thousand people per year.

(C) The population changes by a total of **14** thousand people from $t = 1$ to $t = 4$.

(D) At exactly $t = 4$ years, the population is changing at a rate of **14** thousand people per year.

7 pts

11.

Let $f(x) = \frac{1}{x}$. Calculate the average rate of change of f over each interval. (Rounded answers to 2+ decimal places are fine.)

a.) $[1, 2]$ _____ (1)

b.) $[1, 1.5]$ _____ (1)

c.) $[1, 1.1]$ _____ (1)

d.) $[1, 1.001]$ _____ (1)

e.) As the interval shrinks toward $x = 1$, these values approach... (1)

(A) **1**

(B) **-1**

(C) **0**

(D) **-2**

5 pts

12.

Let $f(x) = x^2 + 2x + 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) $x = 0, h = 1$ _____ (1)

c.) $x = 0, h = 0.1$ _____ (1)

d.) $x = 0, h = 0.01$ _____ (1)

e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

7 pts

13.

Let $f(x) = \frac{-4}{x}$. Use the **limit definition** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ _____ (3)

b.) $f'(2) =$ _____ (2)

c.) Slope of the tangent line to f at $x = -1$ _____ (2)

d.) What does $f'(2) = 1$ mean? (1)

(A) The value of f at $x = 2$ is 1.

(B) The average rate of change of f from $x = 0$ to $x = 2$ is 1.

(C) The slope of the secant line from $x = 2$ to $x = 2 + 1$ is 1.

(D) The instantaneous rate of change of f at $x = 2$ is 1 — the slope of the tangent line there.

8 pts

14. Let $f(x) = x^3 + 2x$. Find the equation of the tangent line to the graph of f at $x = 1$. 7 pts

a.) $f(1)$ _____ (2)

b.) $f'(1)$ _____ (2)

c.) Tangent line _____ (3)

15.

Let $h(t) = -16t^2 + 160t$ be the height, in feet, of a rocket t seconds after launch.

a.) $h'(t) =$ _____ (3)

b.) $h'(1)$ _____ (2)

c.) Interpret $h'(1) = 128$. (1)

(A) At $t = 1$ seconds, the height of the rocket is decreasing at a rate of 128 feet per second.

(B) Over the first 1 seconds, the height of the rocket changed at an average rate of 128 feet per second.

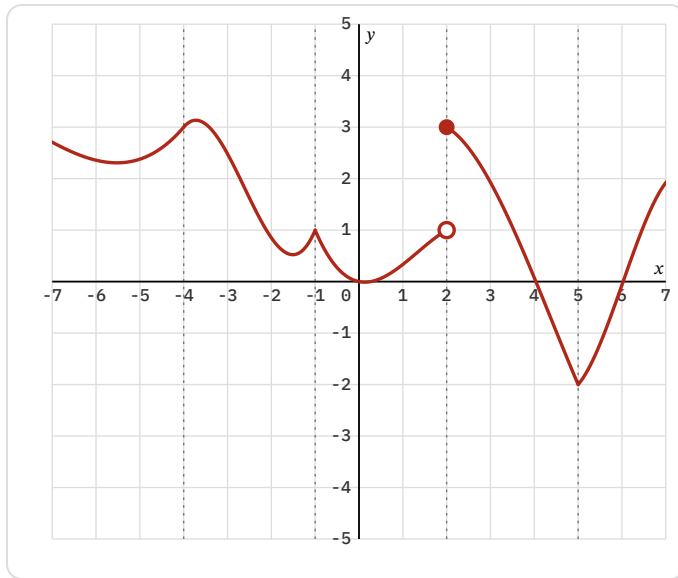
(C) At $t = 1$ seconds, the height of the rocket is increasing at a rate of 128 feet per second.

(D) At $t = 1$ seconds, the height of the rocket is 128 feet.

6 pts

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



a.) At $x = -4$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

b.) At $x = -1$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

c.) At $x = 2$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

d.) At $x = 5$ (1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

4 pts

